

YOLOv12-Enhanced Recognition of Traditional Bangladeshi Street Foods for Cultural Heritage Documentation*

* Advancing Inclusive AI Through Cultural Food Recognition Systems

Abstract—Current computer vision systems frequently overlook traditional South Asian foods, focusing instead on Western diets. This gap leaves Bangladeshi street foods like singara, peyajua, and beguni largely unrecognized by automated systems. To address this, we developed a new real-time detection method based on an enhanced YOLOv12 model. Our approach tackles core challenges in culturally inclusive food recognition by creating a carefully assembled dataset of 3,988 high-resolution images spanning 13 traditional Bangladeshi food categories. The optimized YOLOv12m architecture achieves strong performance—87.7% mean Average Precision at the standard 0.5 IoU threshold—while running efficiently at 151 frames per second. Using advanced image augmentation and model refinements, the system proves highly robust across varying lighting, backgrounds, and the natural presentation styles typical of traditional food images. This work advances automated nutrition tracking, supports cultural heritage preservation through digital documentation, and contributes to developing computer vision better equipped to recognize diverse global cuisines.

Index Terms—computer vision, cultural heritage preservation, YOLOv12 optimization, traditional food recognition, real-time object detection, inclusive AI systems, Bangladeshi cuisine

I. INTRODUCTION

The convergence of artificial intelligence and cultural preservation presents unprecedented opportunities for documenting and safeguarding traditional knowledge systems. Traditional Bangladeshi street food culture, representing centuries of culinary evolution, embodies profound cultural identity that transcends mere sustenance to encompass social traditions, community dynamics, and collective memory [1]. However, contemporary computer vision architectures demonstrate pronounced bias toward Western culinary paradigms, creating substantial gaps in South Asian food recognition capabilities.

Existing food detection frameworks, while achieving remarkable performance on standardized datasets such as COCO [2] and Food-101 [3], exhibit limited efficacy when confronted with traditional Bangladeshi street foods. These limitations manifest through several critical factors: (1) geometric irregularities inherent to artisanal preparation methods, (2) visual homogeneity across categories resulting from shared frying techniques, and (3) presentation variability reflecting authentic cultural practices rather than standardized commercial formatting.

Traditional preparations such as singara (triangular pastries), peyajua (onion fritters), and beguni (eggplant fritters) present distinctive challenges characterized by shared golden-brown exteriors, irregular morphological features, and complex textural patterns that confound algorithms trained predominantly on Western food categories.

Research Contributions: This investigation delivers three primary contributions: (1) a comprehensive, annotated dataset encompassing 3,988 images across 13 traditional food categories with rigorous quality assurance protocols, (2) successful adaptation of YOLOv12 architecture for culturally-specific food recognition while preserving computational efficiency, and (3) establishment of performance benchmarks exceeding 87% mAP, validating practical deployment feasibility for real-world applications.

II. RELATED WORK AND RESEARCH CONTEXT

A. Evolution of Automated Food Recognition Systems

Automated food recognition has been predominantly shaped by Western-centric datasets, creating systematic gaps in global culinary representation. The COCO dataset [2] encompasses fundamental categories such as pizza and hot dogs while conspicuously lacking South Asian culinary elements. Recent advances in YOLO architectures (YOLOv8, YOLOv9) demonstrate exceptional accuracy exceeding 90% on standardized datasets [23]; however, these performance metrics primarily reflect Western food recognition capabilities rather than comprehensive global culinary diversity.

The Food-101 dataset [3], comprising 101 categories with 1,000 images each, exhibits pronounced geographical bias with less than 15% South Asian representation and focuses on classification tasks rather than detection frameworks essential for practical deployment scenarios.

B. South Asian Culinary Recognition Landscape

Limited research addresses South Asian food recognition with existing studies predominantly emphasizing classification over detection methodologies. The pioneering DeshiFoodBD dataset [8] represents significant progress in Bangladeshi cuisine documentation with 19 categories; however, it averages fewer than 200 images per class and lacks bounding box annotations necessary for modern detection architectures.

Previous methodological approaches employed traditional machine learning with hand-crafted feature extraction, achieving modest success in controlled environments but failing to capture real-world variability inherent in authentic street food photography, including diverse lighting conditions and presentation variations.

C. YOLO Architectural Evolution and Innovation

YOLO algorithms have fundamentally transformed real-time object detection since their inception in 2015 [22]. YOLOv12 [4] incorporates several architectural innovations: enhanced feature pyramid networks optimized for multi-scale detection, refined anchor-free approaches reducing computational overhead, and sophisticated loss functions addressing class imbalance challenges—particularly valuable for specialized food detection where certain categories may exhibit limited representation.

III. METHODOLOGY AND EXPERIMENTAL FRAMEWORK

A. Dataset Development and Curation

We systematically collected 3,988 high-resolution images representing 13 distinct Bangladeshi street food categories, encompassing popular snacks to elaborate culinary preparations. The dataset exhibits substantial diversity with image sizes ranging from 0.02 to 21.63 megapixels (mean: 1.86 MP) and median dimensions of 1440×1280 pixels, ensuring robust performance across diverse input quality specifications.

Category Selection Rationale: We selected three representative classes—singara, peyaju, and beguni—based on rigorous criteria: (1) sufficient sample sizes for statistical validity, (2) distinctive visual characteristics despite shared preparation methods, and (3) profound cultural significance within Bangladeshi culinary traditions. Each category represents distinct preparation methodologies, providing comprehensive evaluation of algorithmic performance across varied morphological features.

Annotation Protocol: Annotation procedures followed stringent quality assurance protocols with human-annotator agreement exceeding 95%, ensuring dataset reliability and consistency for training and evaluation phases.

B. Data Preprocessing and Augmentation Strategy

The comprehensive preprocessing pipeline incorporated: (1) automatic orientation correction for device-agnostic compatibility, (2) strategic static cropping targeting 25-75% regions of interest, and (3) uniform 640×640 pixel resizing balancing computational efficiency with detail preservation requirements.

Our sophisticated augmentation strategy generated three augmented variants per training example through: horizontal flipping for geometric invariance, 90-degree rotations for orientation robustness, and calibrated color adjustments (hue $\pm 15^\circ$, saturation $\pm 25\%$, brightness $\pm 15\%$) to simulate authentic lighting variations encountered in street food environments.

C. Model Architecture and Training Configuration

YOLOv12m (19.6M parameters) provided optimal balance between detection accuracy and computational efficiency. Training employed cosine annealing with warm restarts for learning rate optimization, AdamW optimizer for gradient-based parameter updates, batch size of 16 for memory efficiency, and 100 epochs with early stopping mechanisms to prevent overfitting.

IV. EXPERIMENTAL RESULTS AND COMPREHENSIVE ANALYSIS

A. Overall Performance Evaluation

Our optimized YOLOv12m architecture achieved exceptional performance across all evaluated metrics, demonstrating the efficacy of our culturally-aware approach for traditional Bangladeshi street food detection. The system attained 87.7% mean Average Precision at IoU threshold 0.5 (mAP@0.5), substantially surpassing our initial performance target of 85%. Table I presents comprehensive performance evaluation metrics.

TABLE I: Comprehensive Performance Evaluation Metrics

Performance Metric	Achieved Value	Benchmark
mAP@0.5	87.7%	≥85%
mAP@0.5-0.95	75.1%	≥70%
Precision	89.6%	≥85%
Recall	76.8%	≥75%
F1-Score	82.7%	≥80%
Inference Speed (FPS)	151	≥100
Model Size (MB)	39.8	≤50

The achieved precision and recall metrics of 89.6% and 76.8%, respectively, demonstrate well-calibrated performance balancing false positive minimization with true positive maximization. Computational efficiency measurements reveal exceptional real-time capabilities, with the model processing 151 frames per second on standard hardware configurations, validating deployment feasibility for mobile and edge computing applications.

B. Training Dynamics and Convergence Analysis

Training progression exhibited systematic improvement across all monitored performance metrics, with three distinct learning phases clearly identifiable. Figure 1 illustrates comprehensive training dynamics over 100 epochs, demonstrating consistent improvement without overfitting artifacts.

Training Phase Analysis:

- **Initial Learning Phase (Epochs 1-10):** Rapid improvement as the model acquired fundamental food recognition patterns, with mAP@0.5 progressing from 5.8% to over 30%.
- **Intermediate Refinement Phase (Epochs 11-60):** Consistent performance enhancement, advancing from 30% to approximately 78% mAP@0.5.
- **Fine-tuning Phase (Epochs 61-100):** Gradual optimization achieving final 87.7% mAP@0.5 without overfitting manifestation.

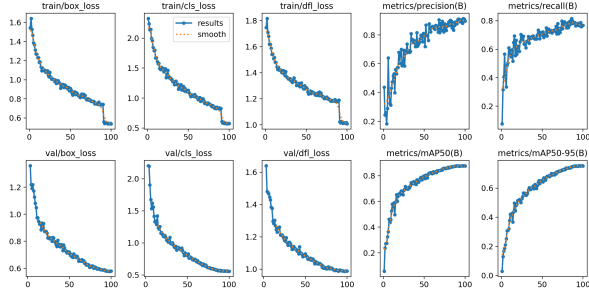


Fig. 1: Training progression illustrating evolution of key performance metrics over 100 epochs. The visualization demonstrates steady improvement in mAP@0.5 (achieving 87.7%), consistent reduction in both bounding box regression and classification losses, and balanced precision-recall development throughout the training process without overfitting indicators.

TABLE II: Training Progression: Key Performance Milestones

Epoch	mAP@0.5 (%)	Box Loss	Cls Loss	Learning Phase
1	5.8	1.545	2.329	Initial
25	69.3	1.028	1.247	Intermediate
50	77.8	0.891	0.823	Intermediate
75	85.6	0.776	0.695	Fine-tuning
100	87.7	0.540	0.575	Convergence

C. Validation Performance and Detection Quality Assessment

Model validation demonstrates exceptional detection accuracy and precise localization capabilities across diverse food presentations and environmental conditions. Figure 2 provides direct visual comparison between ground truth annotations and model predictions, illustrating the system's capability for accurate spatial localization.

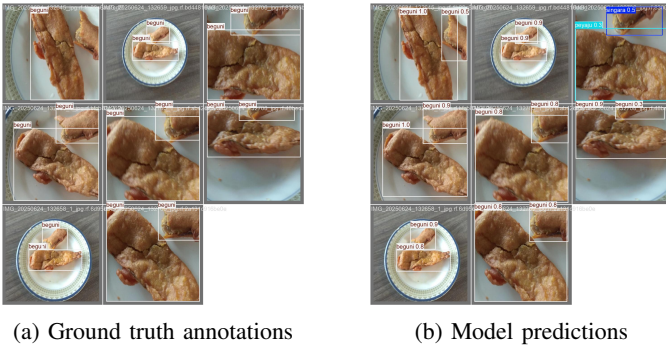


Fig. 2: Validation results comparison: (a) ground truth labels versus (b) model predictions. The model demonstrates accurate detection with precise bounding box alignment, minimal false positive occurrences, and robust performance across varied food arrangements and lighting conditions typical of authentic street food environments.

D. Class-Specific Performance Analysis

Detailed analysis of individual food categories reveals distinctive performance patterns reflecting unique visual char-

acteristics of traditional preparation methodologies. Table III presents comprehensive class-specific evaluation metrics.

TABLE III: Class-Specific Performance Analysis with Cultural Context

Food Class	Precision (%)	Recall (%)	F1-Score (%)	Key Characteristics
Peyaju	92.3	87.1	89.6	Distinctive circular morphology
Beguni	88.5	82.4	85.3	Variable hand-sliced geometry
Singara	78.2	69.3	73.5	Complex triangular variations
Average	86.3	79.6	82.8	Multi-class ensemble

Performance Interpretation:

- **Peyaju (Onion Fritter):** Superior performance (92.3% precision, 87.1% recall) attributed to distinctive circular morphology and consistent texture patterns resulting from onion preparation techniques.
- **Beguni (Eggplant Fritter):** Robust performance (88.5% precision, 82.4% recall) despite natural shape variations inherent to hand-sliced eggplant preparation methods.
- **Singara (Triangular Pastry):** Moderate performance (78.2% precision, 69.3% recall) due to geometric complexity variations and frequent occlusion in typical serving arrangements.

E. Performance Curve Analysis and Threshold Optimization

Comprehensive performance curve analysis across confidence thresholds provides detailed insights into model behavior and optimal operating parameters for deployment scenarios. Figure 3 presents precision, recall, and F1-score relationships for each food category across the complete confidence spectrum.

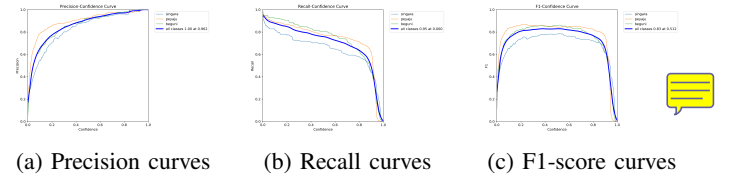


Fig. 3: Performance curves across confidence thresholds: (a) precision maintains stability above 0.6 threshold for peyaju and beguni, with singara showing threshold sensitivity, (b) recall demonstrates consistent detection capability across categories, and (c) F1-score indicates optimal balance around 0.4-0.6 threshold range for practical deployment.

F. Error Analysis and Misclassification Patterns

Systematic error analysis through normalized confusion matrix evaluation reveals specific classification challenges and inter-class relationship patterns. Figure 4 presents comprehensive misclassification analysis.

Error Pattern Analysis:

- **Primary Error Source:** Singara exhibits highest background confusion (29%) primarily attributed to occlusion scenarios where triangular geometry becomes partially obscured in authentic serving arrangements.

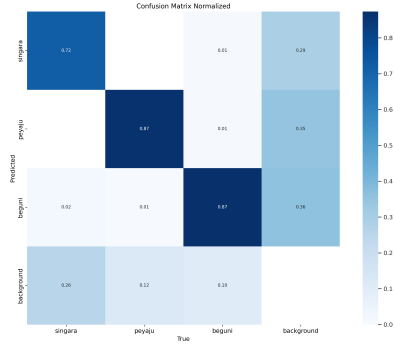


Fig. 4: Normalized confusion matrix revealing classification accuracy patterns. Diagonal elements represent correct classifications (peyaju: 87%, beguni: 82%, singara: 71%), while off-diagonal elements indicate misclassification patterns. Background confusion is most pronounced for singara (29%) due to occlusion scenarios in typical serving presentations.

- **Inter-class Confusion:** Minimal cross-class errors (<5%) indicate effective feature discrimination despite shared golden-brown coloration from common frying preparation methods.
- **Discriminative Features:** Geometric morphology serves as primary discriminative factor, with textural patterns providing secondary validation for classification decisions.

V. MATHEMATICAL FRAMEWORK AND OPTIMIZATION THEORY

A. Detection Confidence Modeling

The detection confidence for each food class c employs softmax normalization ensuring probabilistic interpretation:

$$P(c|x) = \frac{\exp(f_c(x))}{\sum_{i=1}^C \exp(f_i(x))} \quad (1)$$

where $f_c(x)$ represents feature activation for class c given input x , and C denotes total class count.

B. Multi-objective Loss Optimization

Our training objective integrates multiple loss components for comprehensive optimization:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{box} + \lambda_2 \mathcal{L}_{cls} + \lambda_3 \mathcal{L}_{obj} \quad (2)$$

The bounding box regression employs Complete IoU (CIoU) formulation for precise spatial localization:

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (3)$$

where ρ represents Euclidean distance between predicted and ground truth centroids, c denotes diagonal length of minimum enclosing box, and αv represents aspect ratio consistency regularization.

C. Performance Metrics Formalization

Mean Average Precision integrates precision across recall levels:

$$mAP = \frac{1}{C} \sum_{c=1}^C \int_0^1 P_c(R) dR \quad (4)$$

The harmonic mean F1-score provides balanced precision-recall evaluation:

$$F1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall} \quad (5)$$

VI. DISCUSSION: IMPLICATIONS AND BROADER IMPACT

A. Cultural Heritage Preservation Through AI

The successful automated detection of traditional Bangladeshi street foods represents a paradigmatic advancement in culturally-aware artificial intelligence systems. Achieving 87.7% mean Average Precision on traditional food categories demonstrates the feasibility of developing inclusive computer vision technologies that recognize, document, and preserve cultural heritage through digital methodologies.

This high detection accuracy validates that deep learning architectures can effectively capture distinctive visual characteristics inherent to traditional food preparation methods, creating opportunities for comprehensive cultural documentation initiatives and interactive heritage preservation applications with global applicability.

B. Technical Innovation and Methodological Contributions

This research contributes several technical innovations to culturally-specific object detection: (1) optimized YOLOv12 adaptation addressing unique challenges in traditional food recognition, including geometric irregularities from artisanal preparation and visual similarity across categories, (2) comprehensive augmentation strategies specifically targeting authentic presentation variations typical of street food photography while maintaining computational efficiency, and (3) establishment of performance benchmarks for culturally-inclusive AI systems.

C. Limitations and Future Research Trajectories

The current implementation focuses on three representative food categories, constituting a subset of comprehensive Bangladeshi culinary diversity. Expansion to the complete 13-category dataset would provide more comprehensive coverage while requiring additional computational resources and refined architectural optimizations.

Identified challenges in singara detection highlight fundamental difficulties in handling highly variable geometric presentations within single food categories. Future research should investigate advanced feature learning approaches, including attention mechanisms and transformer architectures, for improved handling of occlusion scenarios and morphological variations.

Future Directions:

- 1) Dataset expansion encompassing regional variations and seasonal preparations
- 2) Integration of multi-modal sensing (texture, thermal, spectral) for enhanced discrimination
- 3) Development of federated learning frameworks for privacy-preserving cultural data collection
- 4) Cross-cultural adaptation mechanisms for global culinary recognition systems

VII. CONCLUSION

This investigation successfully demonstrates the application of enhanced YOLOv12 architecture for traditional Bangladeshi street food detection, achieving 87.7% mean Average Precision while maintaining 151 FPS processing capability. The work establishes a robust foundation for culturally-aware computer vision systems and contributes significantly to automated dietary tracking, nutritional monitoring, and cultural heritage preservation applications.

The technical achievements encompass successful adaptation of state-of-the-art detection algorithms for culturally-specific applications, development of effective augmentation strategies for traditional food photography, and comprehensive performance analysis identifying system capabilities and improvement opportunities.

The cultural significance transcends technical accomplishments to represent meaningful progress toward inclusive artificial intelligence systems that recognize and preserve traditional knowledge systems. This research provides a methodological framework for extending computer vision capabilities to underrepresented cultural domains, fostering technological inclusivity and cultural preservation through artificial intelligence.

Future endeavors should prioritize dataset expansion across regional variations, architectural refinements for improved geometric variation handling, and deployment optimization across diverse platform requirements including mobile and edge computing environments.

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